

A relay is an electrically operated switch. It consists of a set of input terminals for a single or multiple control signals, and a set of operating contact terminals. The switch may have any number of contacts in multiple contact forms, such as make contacts, break contacts, or combinations thereof.

It works on the principle of electromagnetic attraction. When the circuit of the relay senses the fault current, it energizes the electromagnetic field which produces a temporary magnetic field. This magnetic field moves the relay armature for opening or closing the connections. The small power relay has only one contact, and the high power relay has two contacts for opening the switch.

Key Features

- Coil Voltage: 5 V DC operation
- High Switching Capacity: Supports up to 10 A at 250 V AC / 30 V DC
- Compact Design: Standard PCB mountable package
- Galvanic Isolation: Provides safe isolation between control circuit and load
- Widely Compatible: Can be driven by microcontrollers (Arduino, STM32, ESP32, etc.) with driver circuitry



1-Channel Relay Module

Technical Specifications

Parameter	Value
Coil Voltage	5 V DC (Nominal)
Coil Resistance	~70 Ω (±10%)
Contact Rating	10 A @ 250 V AC or 10 A @ 30 V DC
Contact Arrangement	SPDT (Single Pole Double Throw)
Mechanical Life	≥10 million operations
Electrical Life	≥100,000 operations

Applications

- **Home automation** — switching lights, fans, or appliances.
- **Industrial control** — motor control, pumps, and machinery.
- **Smart IoT systems** — relay-controlled automation projects.
- **Power electronics** — switching high-power loads safely.